AP Calc Lesson 2.4 Homework

Name ______________________

1. If $f(x) = x^3 + 4\sqrt{x} + \frac{6}{x}$, find $f'(x)$.

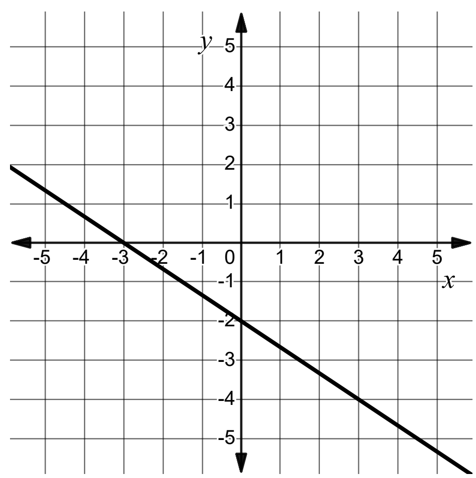
2. Let f be the function given by $f(x) = x^3 + 6x^2 - 15x + 2$.

   a. Find $f'(x)$.

   b. Find the x -coordinate of each point on the curve where f has a horizontal tangent.

3. The graph of $g(x)$ is shown. Let $f(x) = 4x^3 + 2x + 6g(x)$.

   a. Find $f(3)$.



   b. Find $f'(x)$.

   c. Write an equation for the line tangent to f at $x = 3$.

4. The position of a particle moving along the x -axis is given by $s(t) = 8t^{3/2} - 4t + 1$.

   a. Find the average velocity of the particle on the interval $[0, 9]$

   b. Find the instantaneous velocity of the particle at time t.

   c. For what value of t does the average veclocity of the particle on the interval $[0, 9]$ equal the instantaneous velocity of the particle?

5. Let $f(x) = -6x^{2/3} + bx - 7$, where b is a constant.

a. Let $f'(x)$ in terms of b.

b. Evaluate $\lim_{x \to 8} \dfrac{f(x) - f(8)}{x - 8}$

6. The graphs of $f(x) = ax - x^2$ and $g(x) = bx^2 - 2ax + c$ are tangent to each other at the point $(2, 0)$. Findthe values of a, b, and c.

7. Selected values of a polynomial function f and its derivative, f' are given in the table.

x	$f(x)$	$f'(x)$
0	-8	1
2	-18	-7
4	-20	9
6	34	49

The function g is the image of f after a vertical dilation by a factor of 3, a horizontal shift 2 units to the left, and a vertical shift down 5. Find $g'(2)$.

8. Let $f(x) = ax^4 + (a-3)x^3 + ax^2 + a^2x + a$ for some constant a. If the y-intercept of the graph of $y = f'(x)$, the derivative of f, is 16, write a possible equation for $f(x)$.

9. If $f(x) = \dfrac{7}{x^2} + 11x^4$, then $f'(x) =$

A) $\dfrac{7}{2x} + 44x^3$

B) $-14x + 44x^3$

C) $-\dfrac{14}{x^3} + 44x^3$

D) $\dfrac{14}{x^3} + 44x^3$

10. Let f be the function given by $f(x) = \begin{cases} 8\sqrt{x} + 3x^2 & x < 4 \\ x^2 + 18x - 24 & x \geq 4 \end{cases}$. What is $f'(4)$?

A) 26

B) 32

C) 64

D) Nonexistent